ELSEVIER

Materials Science and Engineering A 441 (2006) 26–38

MATERIALS SCIENCE & ENGINEERING A

www.elsevier.com/locate/msea

Oxidation of Nb–Si–Cr–Al *in situ* composites with Mo, Ti and Hf additions

Jie Geng [a], Panos Tsakiropoulos [b,*], Guosheng Shao [a]

[a] *Metallurgy Research Group, School of Engineering, University of Surrey, Guildford, Surrey GU2 7XH, UK*

[b] *Department of Engineering Materials, Sir Robert Hadfield Building, The University of Sheffield, Mappin Street, Sheffield S1 3JD, UK*

Received 1 July 2005; received in revised form 1 February 2006; accepted 1 August 2006

Abstract

The effects of Mo, Ti and/or Hf additions on the isothermal oxidation behaviour of Nb–Si–Cr–Al-based *in situ* composites in static air at 800 °C and 1200 °C were studied for the as-cast and heat-treated materials. After heat treatment at 1500 °C, the microstructures of all the alloys consisted only of the niobium solid solution (Nb_{ss}) and Nb_5Si_3 phases. The addition of Ti and the decrease of the Mo concentration to 2 at.% improved the oxidation resistance of the alloys at 800 °C dramatically. The Hf addition had no significant effect on the oxidation behaviour of Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf (at.%) at 800 °C. The oxidation resistance of the alloys was found to be sensitive to the volume fraction of Nb_{ss}. The coarsening and the increase of the volume fraction of the Nb_{ss} phase in the heat-treated alloys were mainly responsible for the degradation of their oxidation resistance. Preferential attack of the Nb_{ss} phase was observed in all the alloys. Pesting oxidation behaviour was exhibited at 800 °C by the Nb–18Si–5Al–5Cr–5Mo (as-cast), Nb–24Ti–18Si–5Al–5Cr–5Mo (as-cast), Nb–24Ti–18Si–5Al–5Cr–2Mo (heat-treated) and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf (heat-treated) alloys. Pesting of the alloys occurred due to a combination of the presence of elements that oxidised rapidly and the restricted deformation capability of the scales. Spallation of oxide scales during cooling was observed for the as-cast and heat-treated Nb–24Ti–18Si–5Al–5Cr–2Mo and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf alloys at 1200 °C. In the diffusion zones formed in the alloys without Hf addition, the 5–3 silicide was not oxidised. The $TiNbO_4$, $TiNb_2O_7$, $Ti_2Nb_{10}O_{29}$ and silicon oxide were present in the scales formed on the Nb–24Ti–18Si–5Al–5Cr–2Mo and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf alloys at 1200 °C and Hf oxide was also present in the latter alloy. At 1200 °C, in the Ti containing alloys, large Ti oxide particles formed at the Nb_{ss} and 5–3 silicide interface and fine Ti oxide particles formed inside the Nb_{ss}. Needle-like Hf oxide formed inside the 5–3 silicide in the heat-treated Hf containing alloy, which resulted in the degradation of its oxidation resistance. No protective oxide scale was formed on any of the alloys studied in this work.
© 2006 Elsevier B.V. All rights reserved.

Keywords: Silicides; Various; Oxidation; Electron-probe microanalysis

1. Introduction

In the past 30 years, the temperatures experienced by high-pressure turbine blades have increased by almost 125 °C and their operation temperatures have approached 1150 °C, which is essentially close to the limit for the Ni-based superalloys [1–3], as further increases in the temperature capability of these materials will be very difficult to achieve due to their relative low melting points (∼1350 °C). Thus, high temperature structural materials that can survive above the present temperature limit are desired. Niobium silicide-based *in situ* composites are Nb–Si based alloys with additions of transition and refractory metals and have microstructures that comprise of Nb solid solution and 5–3 silicides, and/or 3–1 silicides and Laves phases. These alloys have been shown to have great potential as alternative materials to Ni-based superalloys in terms of their low density, high melting points and high strength at high temperatures [4–7]. However, their poor oxidation behaviour at intermediate and high temperatures is a major barrier to their use in high-temperature applications [1,7–9].

The literature on microstructure-property studies and phase equilibrium data for Nb silicide-based *in situ* composites is significantly more extensive than the literature on their oxidation behaviour. Previous work has shown that the oxidation resistance of the Nb silicide-based *in situ* composites can be improved significantly by additions such as Cr, Al, Ti and Hf [1,7–9]. Germanium and B additions, as a partial replacement of Si, also improve the oxidation behaviour of Nb silicide-based *in situ*

* Corresponding author. Fax: +44 1142 225943.
E-mail address: p.tsakiropoulos@sheffield.ac.uk (P. Tsakiropoulos).

0921-5093/$ – see front matter © 2006 Elsevier B.V. All rights reserved.
doi:10.1016/j.msea.2006.08.093

composites [2,7,8]. Unfortunately, the level of alloying required for the improvement of oxidation resistance could sacrifice the room temperature fracture toughness and high temperature strength of the composites [1,7–11]. Further investigations are thus needed to achieve the required balance of properties.

Niobium has a high solubility for oxygen, its silicides suffer from pest oxidation at intermediate temperatures and in Nb alloys local equilibrium between the oxide/s and the substrate cannot be established. The poor oxidation behaviour of Nb silicide-based *in situ* composites as well as their susceptibility to interstitial contamination are key issues that alloy design and development must address.

In this paper we present the results of a systematic investigation of the effects of Mo, Ti and Hf additions in Nb–Si–Cr–Al *in situ* composites on oxidation behaviour in static air at 800 °C and 1200 °C.

2. Experimental

The base composition Nb–18Si–5Cr–5Al was selected, as it provides a basis for developing microstructures consisting of the niobium solid solution (Nb_{ss}), Nb_3Si, Nb_5Si_3 and Cr-rich Laves phases, thus allowing study of the role of combinations of these phases in the oxidation behaviour of Nb silicide-based *in situ* composites in the as-cast (AC) and heat-treated (HT) conditions. In the latter materials, tensile ductility and fracture resistance at ambient temperature are provided by the Nb_{ss}, the Si is used for oxidation and creep resistance, the Cr and Al for oxidation resistance, and Ti and Hf are added to improve fracture behaviour and oxidation resistance.

The Nb–18Si–5Al–5Cr–5Mo (JG1, all compositions are given in atomic percent in this paper), Nb–24Ti–18Si–5Al–5Cr–5Mo (JG2), Nb–24Ti–18Si–5Al–5Cr–2Mo (JG3) and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf (JG4) alloys were selected to study the effects of Ti, Mo and Hf additions on their oxidation in static air at 800 °C and 1200 °C. The Si content of all the alloys, which controls the volume fraction of the silicide/s, was set to 18 at.%, since this is reported to be the optimum concentration for creep behaviour [12]. Both Al and Cr were kept at 5 at.%, as at this alloying levels the formation of Laves phase might be possible in the alloy and the ductile brittle transition temperature (DBTT) of the solid solution matrix is not expected to increase above −50 °C, as both Cr and Al increase the DBTT of Nb with Al having the strongest effect [13]. The Ti addition in all the alloys was set at 24 at.% in order to maintain eutectic temperature above 1700 °C and to avoid the formation of the *hP*16 Ti_5Si_3 phase, which has been reported to be detrimental to the creep-rupture strength of the composites [14]. Molybdenum and Hf have a significant strengthening effect for Nb_{ss} at elevated temperatures ($T \geq 1000$ °C), Mo has a positive effect and Hf has a marginally negative effect on the creep strength of Nb [15]. The concentrations of Mo and Hf were not increased above 5 at.%, since at these levels a good balance of strength of Nb–Si base alloys at ambient and high temperature has been reported [16] and Hf, by being kept below 8 at.%, is not expected to have a strong negative effect on creep strength [14].

Ingots were prepared from high purity elements using arc-melting under an argon atmosphere in a water-cooled copper hearth using a non-consumable tungsten electrode. The alloys were remelted five times in order to ensure homogeneity. Samples for heat treatment were wrapped in Ta foil and heat-treated at 1500 °C for 100 h or 200 h in flowing argon, which was first gettered over Ti chips. All samples were furnace cooled. The actual chemical compositions of the alloys, which were determined by electron-probe microanalysis (EPMA), are given in the Table 1.

Study of the oxidation behaviour of the alloys was done by thermo–gravimetric analysis (TGA) using Stanton Redcroft thermo–balances in static air for 50–100 h. The specimens (4 mm × 4 mm × 9 mm, total surface area of ∼1.76 cm^2) were polished to 800 grit SiC paper and cleaned in methanol. Isothermal oxidation tests of the rectangular shape samples were undertaken at 800 °C and 1200 °C. Oxidation specimens were selected from the central regions of the ingots and from the bulk of the heat-treated alloys, in the latter case to ensure that TiN was not present in the microstructure [17,18].

The phases present in the alloys and the oxide scales were identified by X-ray diffraction (XRD) using a Philips diffractometer with monochromatic CuKα radiation (λ = 1.540562 Å). The microstructures were observed by scanning electron microscopy (SEM) and the chemical compositions of the constituent phases were analysed by EPMA using a JEOL JXA 8600 super-probe equipped with EDX and WDX analysis facilities.

EPMA was operated using a 15 kV accelerating voltage, under which the spatial resolution was about 1 μm. Analysis was only performed on particles of diameter >5 μm. High purity Nb,

Table 1
Actual chemical compositions (at.%) and volume fractions (%) of Nb_{ss} of the alloys studied in this work

Alloy	Nb	Ti	Si	Cr	Al	Mo	Hf	Condition	Volume fraction of Nb_{ss} (%)
JG1-AC	67.8	–	17.7	4.8	4.5	5.2	–	As-cast	40.5
JG1-HT	67.2	–	18.1	4.6	4.7	5.4	–	As-cast + 1500 °C/100 h	48.9
JG1-HT2	68.0	–	18.5	4.7	4.7	4.1	–	As-cast + 1500 °C/200 h	51.9
JG2-AC	43.1	24.3	18.2	4.7	4.8	4.9	–	As-cast	34.2
JG2-HT	42.2	25.3	18.2	5.0	4.8	4.5	–	As-cast + 1500 °C/100 h	45.5
JG3-AC	45.1	25.4	17.5	5.2	4.8	2.0	–	As-cast	35.9
JG3-HT	46.0	24.8	17.9	5.1	4.6	1.6	–	As-cast + 1500 °C/100 h	47.2
JG4-AC	41.2	23.6	18.9	4.5	4.6	2.0	5.2	As-cast	39.1[a]
JG4-HT	41.4	24.2	18.2	4.6	4.8	1.7	5.1	As-cast + 1500 °C/100 h	55.4[a]

[a] The measured value may be a little higher than the actual one as under BSE condition the contrast between Nb_{ss} and Hf-rich Nb_5Si_3 was almost the same.

Ti, Si, Cr, Mo, Hf and stoichiometric Al_2O_3 (for Al and O) were used as standards and ZAF iterative corrections were used to calculate the wt.% compositions from measured X-ray intensities. The errors in the Nb, Ti, Si, Cr, Mo, Hf, Al and O measurements were ~0.8%, 0.4%, 0.2%, 0.3%, 0.7%, 0.6%, 0.1% and 0.9%, respectively.

3. Results

3.1. The microstructures of the alloys

The four alloys have been classified as hyper-eutectic Nb silicide-based *in situ* composites as the βNb_5Si_3 phase was present in their as solidified microstructures [17,18]. The Nb_3Si silicide was formed in the JG1-AC, JG2-AC and JG3-AC alloys but not in JG4-AC. Both JG1-AC and JG4-AC contained a small volume fraction of Si-containing C14–Cr_2Nb Laves phase, which was absent in JG2-AC and JG3-AC [17,18]. In JG2-AC, JG3-AC and JG4-AC there were Ti and/or Hf-rich regions in the Nb_5Si_3. After heat treatment at 1500 °C for 100 h or 200 h, the microstructures of all the alloys consisted of a continuous network of coarsened solid solution Nb_{ss} within which the coarsened 5–3 silicide phase was distributed, in other words, the microstructures of all the heat-treated alloys that were oxidised at 800 °C and 1200 °C consisted of the same two phases but with different compositions for each alloy [17,18]. The Ti and/or Hf-rich regions in the 5–3 silicide, which developed due to the partitioning behaviour of these elements, were still present in the JG2-HT, JG3-HT and JG4-HT alloys. The data in Table 1 shows that equilibration of the microstructure of each alloy as a result of heat treatment was accompanied by an increase in the volume fraction of Nb_{ss}. The microstructures of the as-cast and heat-treated alloys are discussed in detail elsewhere [17,18].

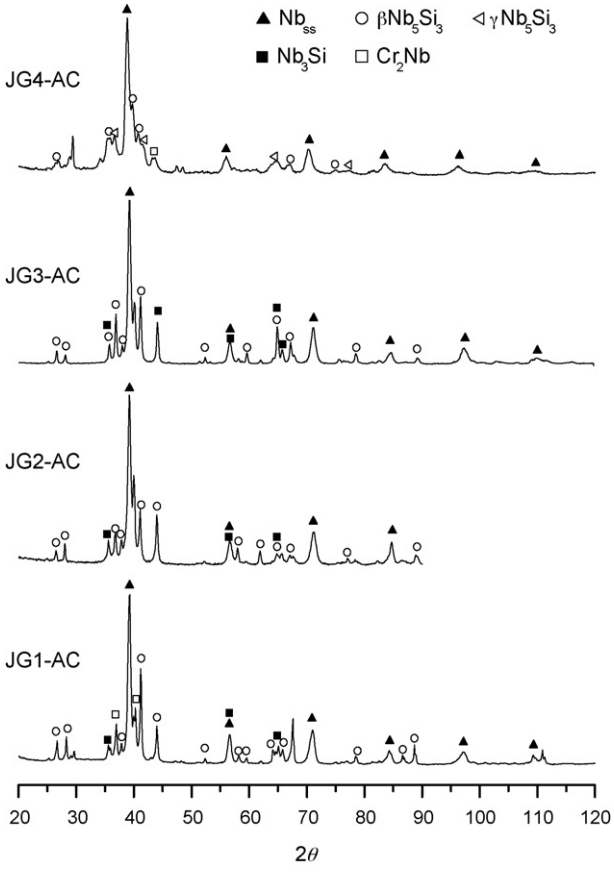


Fig. 1. X-ray diffractograms of the as-cast alloys studied in this work.

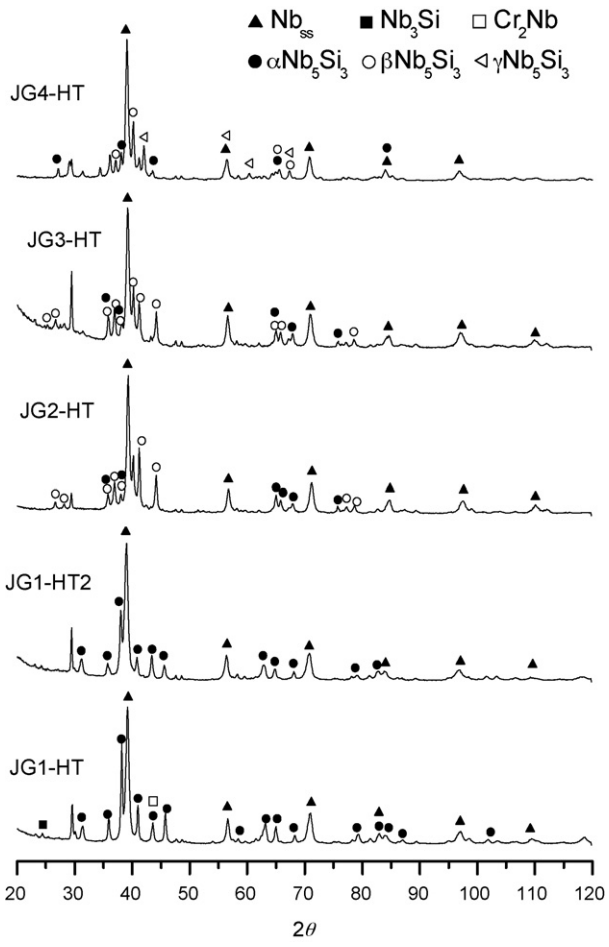


Fig. 2. X-ray diffractograms of the heat-treated alloys studied in this work.

The X-ray diffractograms of all the as-cast and heat-treated alloys are presented in Figs. 1 and 2. Individual phases and their crystal structures were identified by matching the characteristic XRD peaks against JCPDS data. The Nb_5Si_3, which formed in all the as-cast alloys, was the tetragonal $D8_m$ W_5Si_3-type βNb_5Si_3. In the alloy with Hf addition (JG4), the hexagonal γNb_5Si_3 (metastable 5–3 silicide stabilised by interstitials) phase was also formed. After heat treatment at 1500 °C for 100 h, the βNb_5Si_3 phase had completely transformed to the tetragonal $D8_l$ αNb_5Si_3 in JG1, but this transformation was not completed in the alloys with 24 at.% Ti addition (JG2, JG3, JG4). Comparison of JG1-HT, the microstructure of which contained a small volume fraction of Nb_3Si after the heat treatment at 1500 °C for 100 h, with the alloys JG2-HT, JG3-HT and JG4-HT, shows that the addition of Ti in the latter three alloys had enhanced the eutectoid decomposition of Nb_3Si to niobium solid solution (Nb_{ss}) and αNb_5Si_3 considerably. The hexagonal Ti_5Si_3 phase did not form in any of the alloys studied in this work.

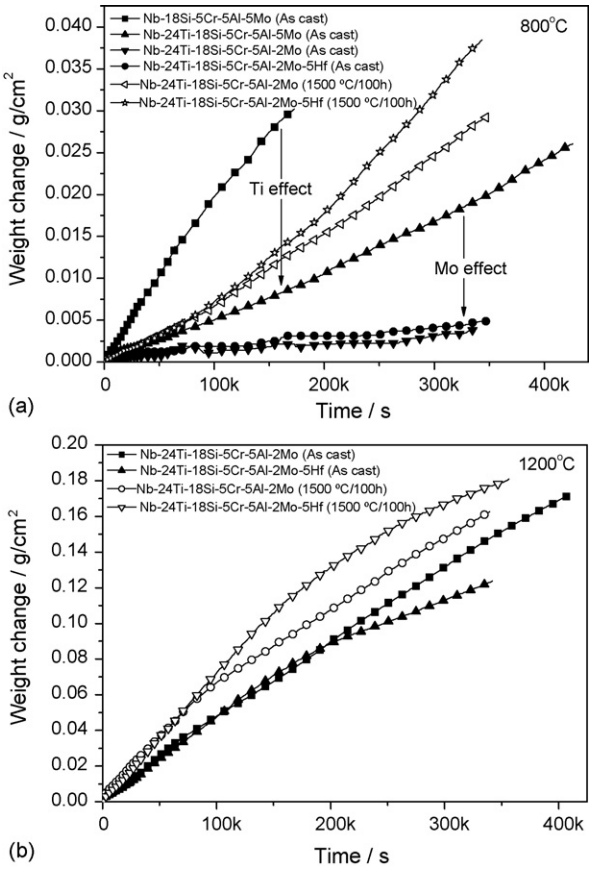


Fig. 3. Isothermal oxidation behaviour of selected alloys in static air at: (a) 800 °C and (b) 1200 °C.

3.2. Oxidation kinetics

The oxidation rates of all the alloys were measured at 800 °C and 1200 °C in static air. Results of the thermo-gravimetric studies are shown in Fig. 3, where the weight change (g/cm^2) is plotted as a function of time (s).

At 800 °C, the isothermal oxidation tests showed that the oxidation kinetics of JG1-AC, JG2-AC, JG3-HT and JG4-HT were linear. Table 2 shows the linear oxidation rate constant k_l for each alloy, which was calculated by using the equation [19]

$$\frac{\Delta m}{A} = k_l t \quad (1)$$

where Δm is the weight change of the specimen, A is the surface area of the specimen and t is the exposure time. The weight changes of the JG3-AC and JG4-AC at 800 °C followed parabolic oxidation kinetics. The parabolic oxidation rate constants k_p for these alloys were calculated by using the equation [19]

$$\left(\frac{\Delta m}{A}\right)^2 = k_p t \quad (2)$$

The parabolic oxidation rate constants of the alloys JG3-AC and JG4-AC at 800 °C are given in Table 2.

The as-cast Nb–18Si–5Al–5Cr–5Mo alloy (JG1-AC), which exhibited the highest weight gain, showed very poor oxidation behaviour at 800 °C. The addition of titanium (alloys JG2, JG3 and JG4) increased the oxidation resistance dramatically at 800 °C. The Mo concentration had also a significant effect on the oxidation behaviour. In the presence of 24 at.% Ti, the alloys with lower Mo content showed better oxidation resistance. However, the Hf addition had no significant effect on the oxidation behaviour of the JG4 alloy at 800 °C. As seen in Fig. 3a, the weight gain of JG4-AC (with Hf addition) was slightly higher than the JG3-AC (no Hf present). The heat-treated alloys JG3-HT and JG4-HT showed poor oxidation resistance at 800 °C. Their weight changes were approximately seven times larger than those of the as-cast alloys with the same composition.

Exposed to static air at 1200 °C, the JG3-AC and JG4-AC alloys showed better oxidation resistance than the heat-treated ones, as shown in Fig. 3b. The alloys JG3 and JG4 showed near identical oxidation behaviour at the initial linear oxidation stage, for ~56 h for the as-cast alloys and ~24 h for the heat-treated alloys. The addition of Hf decreased the oxidation rate of JG4-AC whilst increased the oxidation rate of JG4-HT. The oxidation of JG3-HT, JG4-AC and JG4-HT could be divided in two main stages characterised by: (1) an initial linear oxidation rate, which was followed by (2) a linear stage with lower oxidation rate for JG3-HT and JG4-AC or a parabolic oxidation stage for JG4-HT. The oxidation rate constants of the alloys are given in Table 2.

3.3. Oxide scale morphology and microstructure

3.3.1. Oxidation at 800 °C

At 800 °C, the alloy JG1-AC exhibited catastrophic oxidation behaviour. Exposed to air for 48 h, the rectangular sample of JG1-AC had disintegrated into powder completely, indicating that pest oxidation had occurred. Partial pest oxidation behaviour was exhibited by the JG2-AC, JG3-HT and JG4-HT alloys exposed to air for 48 to ~95 h. The outer parts of

Table 2
The oxidation rate constants of the alloys at 800 °C and 1200 °C

Alloy	800 °C		1200 °C	
	k_l (g cm^{-2} s^{-1})	k_p (g^2 cm^{-4} s^{-1})	k_l (g cm^{-2} s^{-1})	k_p (g^2 cm^{-4} s^{-1})
JG1-AC	1.9×10^{-7}	–	–	–
JG2-AC	5.7×10^{-8}	–	–	–
JG3-AC	–	2.5×10^{-11}	4.4×10^{-7}	–
JG3-HT	8.0×10^{-8}	–	6.7×10^{-7} (0–24 h), 4.1×10^{-7} (>24 h)	–
JG4-AC	–	4.7×10^{-11}	4.4×10^{-7} (0–56 h), 2.4×10^{-7} (>56 h)	–
JG4-HT	1.0×10^{-7}	–	6.7×10^{-7} (0–24 h)	1.1×10^{-7} (>24 h)

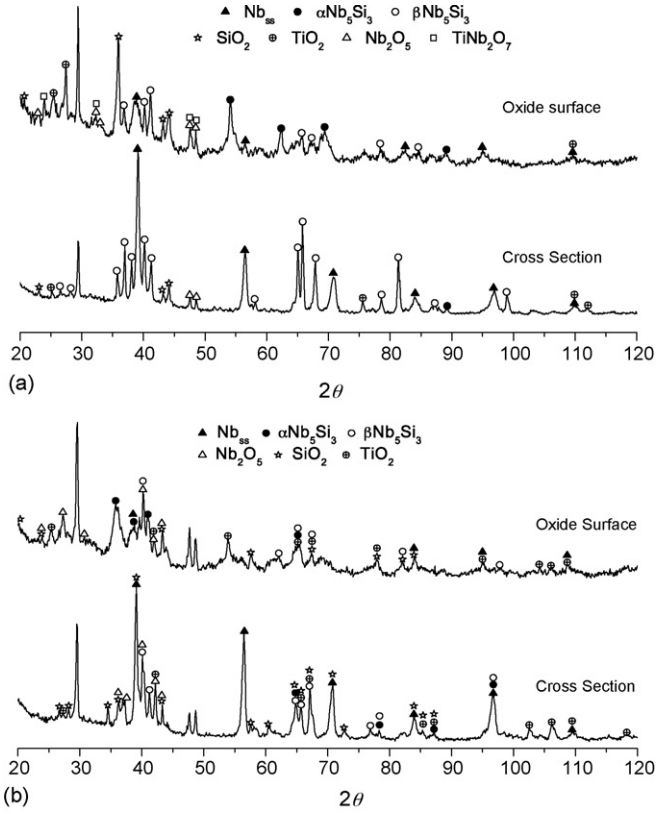


Fig. 4. X-ray diffractograms of the surfaces and cross sections of the: (a) JG3-AC and (b) JG4-AC after oxidation at 800 °C. The unmarked peaks correspond to the specimen holder of the X-ray diffractometer.

the oxidised specimens disintegrated into powder during the isothermal oxidation tests. The surface regions of the remaining parts of the oxidised specimens exhibited cracks parallel to the surface, as was the case for the oxidised JG3-AC and JG4-AC alloys, for which, no spallation of oxide scales was observed at 800 °C.

Fig. 4 shows the X-ray diffractograms of the surfaces and cross sections of the JG3-AC and JG4-AC after oxidation at 800 °C. The XRD data and SEM observations of the scales formed on the oxidised JG3-AC and JG4-AC alloys revealed that the scale microstructures were similar. A typical BSE image of a cross section of the oxidised JG4-AC alloy (800 °C/95 h) is shown in Fig. 5 and the compositions of the phases present in this alloy are listed in Table 3. A non-homogeneous oxide scale was formed. Large oxide particles exhibiting black contrast in

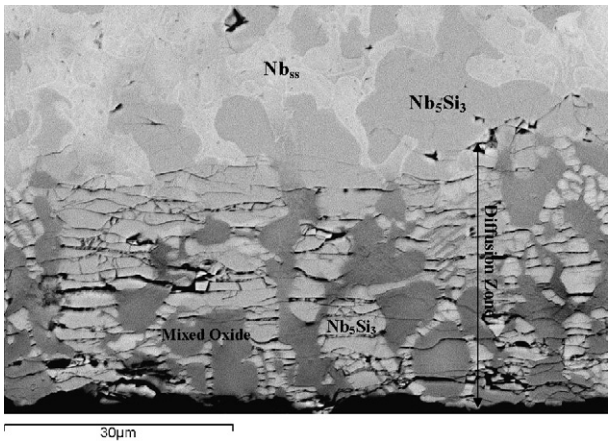


Fig. 5. BSE image of the cross section of JG4-AC after oxidation at 800 °C for 100 h.

the SEM were observed near the surface. These were a complex mixture of Nb and Ti oxides, as confirmed by EPMA (Table 3) and XRD (Fig. 4). The 5–3 silicide grains in the near surface region of the oxidised JG3-AC and JG4-AC exhibited severe cracking, with the cracks in the 5–3 silicide and the oxidised region being parallel to the surface. According to the EPMA results (Table 3), in the oxygen diffusion zone, there was no oxygen dissolved in the 5–3 silicide. HfO_2 was not formed in the diffusion zone of JG4-AC. This would suggest that in the presence of Ti, Al, Cr and Mo, the Hf addition does not change the diffusion zone structure in Nb silicide-based *in situ* composites at intermediate temperatures.

Fig. 6 shows cross sections of the remaining parts of the oxidised JG3-HT and JG4-HT alloys. The compositions of phases present in the oxide scale of both alloys are listed in Table 4. Beneath a highly cracked and porous scale that exhibits black contrast in Fig. 6a and b, there are cracks parallel to the surface of the specimen, as for the oxidised JG3-AC and JG4-AC. Remnants of the 5–3 silicide in the black contrast scale, which are indicated by arrows in Fig. 6b, contained ~9.6 at.% oxygen. The scales that remained adherent to the JG3-HT and JG4-HT alloys were a complex mixture of Nb and Ti oxides (Table 4). The EMPA results indicated that oxygen was dissolved in the niobium solid solution of the alloy under the oxide scale. It is therefore concluded that the oxidation of JG3-HT and JG4-HT occurred by inward oxygen anion diffusion.

Table 3
Compositions (at.%) of the phases present in the JG3-AC and JG4-AC after oxidation at 800 °C

Alloy	Phase	O	Al	Si	Ti	Cr	Nb	Mo	Hf
JG3-AC	Mixed oxide	33.4	4.2	1.6	19.1	6.2	33.5	2.0	–
	Nb_5Si_3	0.0	3.6	31.8	20.1	1.5	43.0	0.0	–
	Nb_{ss} in the bulk	0.0	5.9	1.9	28.7	9.8	50.7	3.1	–
JG4-AC	Mixed oxide	36.3	4.2	1.1	17.1	6.4	30.9	2.0	1.8
	Nb_5Si_3	0.0	3.0	33.9	17.2	0.7	40.7	0.3	4.2
	Nb_{ss} in the bulk	0.0	6.3	2.2	26.5	7.3	50.6	3.9	3.3

The accuracy of oxygen analysis was about ±4.0 at.%.

Table 4
Compositions (at.%) of the phases present in the oxide scale of oxidised JG3-HT and JG4-HT after oxidation at 800 °C

Alloy	Phase	O	Al	Si	Ti	Cr	Nb	Mo	Hf
JG3-HT	Nb-rich oxide	68.4	0.6	10.0	3.7	0.4	16.8	0.0	–
	Ti oxide	67.5	0.0	0.0	29.9	0.0	2.6	0.0	–
JG4-HT	Nb-rich oxide	69.2	2.1	0.2	6.5	3.3	17.7	1.1	0.0
	Ti oxide	66.9	0.0	0.5	29.7	0.0	1.5	0.0	1.4
	Oxygen containing Nb_5Si_3	9.6	3.4	31.2	20.5	1.1	26.5	0.0	7.8

The accuracy of oxygen analysis was about ±4.0 at.%.

3.3.2. Oxidation at 1200 °C

At 1200 °C, spallation of oxide scales during cooling was observed both for the as-cast and heat-treated JG3 and JG4 alloys. Fig. 7 shows the X-ray diffractograms of cross sections of the oxide scale and bulk material of the oxidised JG3-AC and JG4-AC at 1200 °C. The XRD data and SEM observations indicated that the microstructures and morphology of the oxide scales formed on JG3 and JG4 were similar. Fig. 8 shows the secondary electron (SE) and BS images of the porous oxide scale formed on JG4-HT. The high concentration of pores resulted in the weak adherence of the oxide scale and promoted the oxygen diffusion through the scale.

According to the EPMA results (Table 5) three oxides, namely the $Ti_2Nb_{10}O_{29}$ (light phase), $TiNbO_4$ (grey phase) and silicon oxide (black phase), were present in all the scales formed on both JG3 and JG4, and Hf-rich mixed oxide (white phase) was also present in the oxide scale formed on JG4. The accurate determination of the oxide composition by EPMA was not easy, due to the porous nature of the oxide scales and the fine particle size of some oxide constituents. In the oxide scale, the accuracy of oxygen analysis was about ±5.2 at.%. The analysis revealed that the ratio of Si to O in the black silicon oxide phase was close to the stoichiometry of SiO_2. The Mo was not detected in the oxide scales by EPMA.

The XRD data (Fig. 7) confirmed the presence of the phases determined by EPMA but would also suggest the presence of other oxides. Indeed, according to the XRD results, the phases present in the oxide scales formed on JG3-AC and JG4-AC at 1200 °C consisted of $Ti_2Nb_{10}O_{29}$, Ti_2NbO_7, SiO_2, TiO_2 and

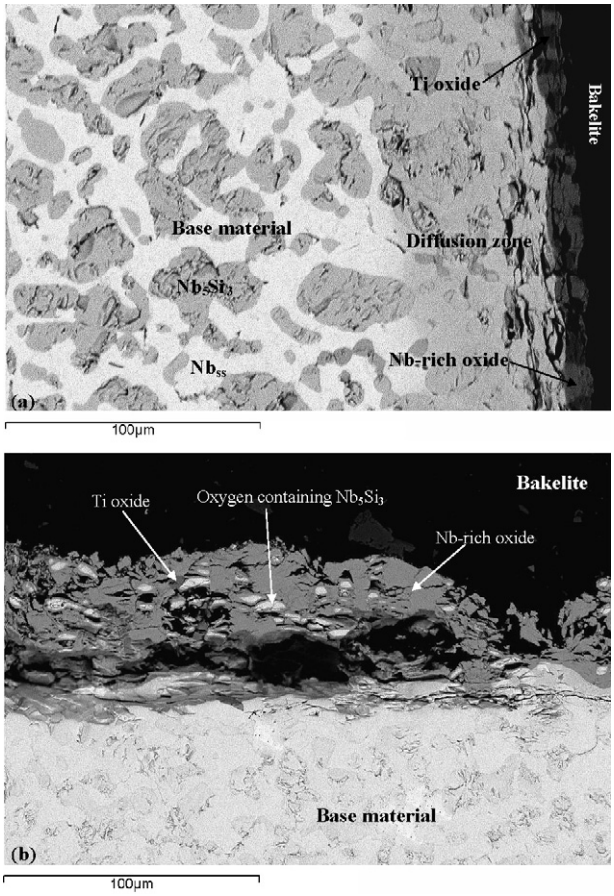


Fig. 6. BSE images of the microstructures of cross sections of the: (a) JG3-HT and (b) JG4-HT after oxidation at 800 °C.

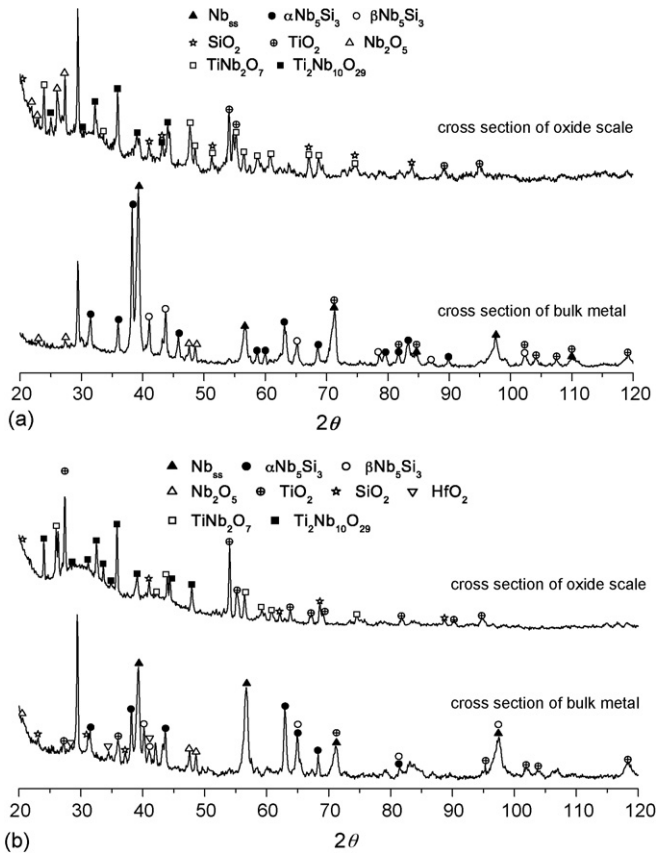


Fig. 7. X-ray diffractograms of cross sections of the oxide scale and bulk material of the oxidized: (a) JG3-AC and (b) JG4-AC at 1200 °C. The unmarked peaks correspond to the specimen holder of the X-ray diffractometer.

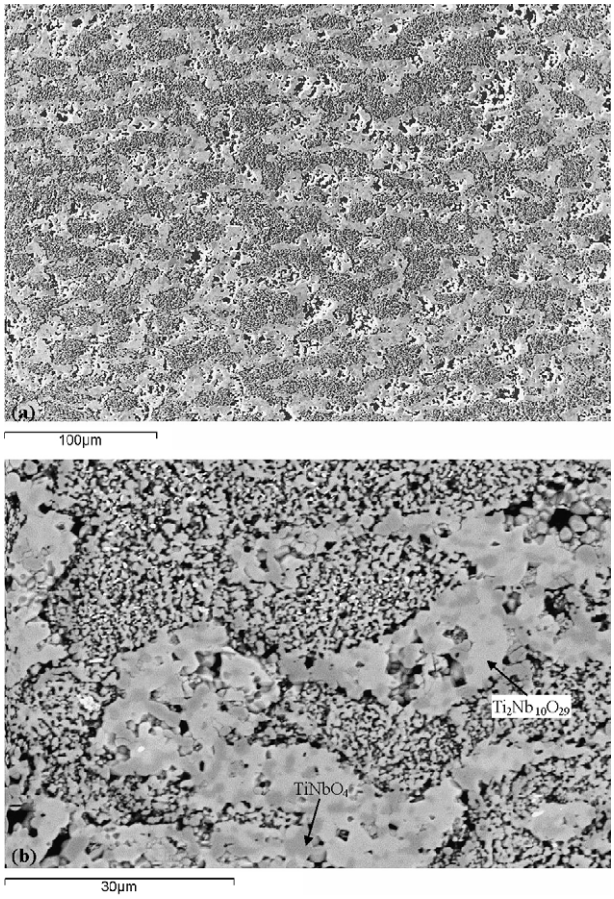


Fig. 8. SEM images of the cross section of the oxide scale formed on JG4-HT at 1200 °C: (a) secondary electron image (300×) and (b) BSE image (1500×). The black phase is silicon oxide and the white phase is Hf-rich oxide.

Nb_2O_5 for the former alloy (Fig. 7a) and $Ti_2Nb_{10}O_{29}$, Ti_2NbO_7, SiO_2, TiO_2 for the latter alloy for which Nb_2O_5 and HfO_2 were also present together with the previous oxides below the oxidised surface (Fig. 7b).

The SEM observations of cross sections of the oxidised specimens revealed that the core of the specimen had the same microstructure as the as-cast and heat-treated alloy [17,18]. However, there was formation of diffusion zones between the surface and the core of the specimen. Internal oxidation was observed in both the JG3 and JG4 alloys at this temperature. Figs. 9–11 show BSE images of cross sections (including the core of the specimen) of the JG3-HT, JG4-AC and JG4-HT alloys oxidised at 1200 °C.

In the oxidised JG3-HT (Fig. 9a), the diffusion zone exhibited two different microstructures, which are classified as zones I and II. In the diffusion zone I (see Fig. 9b), the Nb solid solution was converted to a complex mixture of TiO_x (black phase) and (Nb, Ti, Cr, Mo, Al)O_x (white phase). Large TiO_x particles were also found at the interface between the Nb_{ss} and 5–3 silicide. No oxygen was dissolved in the 5–3 silicide phase. In the diffusion zone II (Fig. 9c) and the core of the specimen (Fig. 9d), the microstructure consisted of three phases, namely Nb_{ss}, Nb_5Si_3 and titanium oxide. However, the volume fraction of titanium oxide in the bulk was significantly smaller than that in the diffusion zones. The titanium oxide formed either at the interface between the Nb_{ss} and 5–3 silicide or inside the Nb_{ss}. In the diffusion zone II, a low concentration of oxygen (∼6 at.%) was dissolved in the niobium solid solution while the 5–3 silicide in this region was not oxidised. No titanium segregation in the 5–3 silicide phase was found in the oxidised samples, compared with JG3-AC and JG3-HT, in which Ti-rich 5–3 silicide phase was present [17].

The microstructures of the JG4 alloy after isothermal oxidation at 1200 °C are shown in Fig. 10 for the as-cast and Fig. 11 for the heat-treated conditions, respectively. In the outermost diffusion zone I (Fig. 10b and Fig. 11b), the TiO_x (black phase) was formed either inside the niobium solid solution (light grey phase in Fig. 10b and Fig. 11b) or at the interface between the Nb_{ss} and 5–3 silicide. About 8 at.% oxygen was dissolved in the Nb_{ss}. Faceted/bulky HfO_2 formed at the interface between the Nb_{ss} and 5–3 silicide and needle-like HfO_2 formed inside the

Table 5
Compositions (at.%) of the phases present in the oxide scales formed on the alloys JG3 and JG4 after oxidation at 1200 °C

Alloy	Phase	O	Al	Si	Ti	Cr	Nb	Mo	Hf
JG3-AC	$Ti_2Nb_{10}O_{29}$	76.4	0.7	0.0	4.1	0.0	18.8	0.0	–
	$TiNbO_4$	75.1	3.4	0.0	10.4	3.3	7.9	0.0	–
	SiO_2	73.0	0.0	26.1	0.2	0.0	0.8	0.0	–
JG3-HT	$Ti_2Nb_{10}O_{29}$	74.4	1.0	0.0	5.3	0.2	19.1	0.0	–
	$TiNbO_4$	73.0	3.7	0.1	11.4	3.3	8.6	0.0	–
	Si_2O_5	72.2	0.7	23.5	1.6	0.7	1.3	0.0	–
JG4-AC	$Ti_2Nb_{10}O_{29}$	69.9	1.5	0.0	7.6	0.7	18.5	0.0	1.8
	$TiNbO_4$	69.5	3.2	0.0	13.7	3.7	7.8	0.0	2.2
	SiO_2	68.8	0.4	24.9	2.7	0.5	2.7	0.0	0.0
	Hf-rich oxide	70.7	1.1	7.1	5.4	1.1	6.8	0.0	7.9
JG4-HT	$Ti_2Nb_{10}O_{29}$	72.9	1.2	0.0	5.8	0.4	18.9	0.0	0.7
	$TiNbO_4$	71.2	3.3	0.3	12.2	3.6	8.0	0.0	1.4
	SiO_2	71.6	0.4	23.9	1.9	0.4	1.8	0.0	0.0
	Hf-rich oxide	69.1	1.6	9.6	5.2	1.8	5.7	0.0	7.0

The accuracy of oxygen analysis was about ±5.2 at.%.

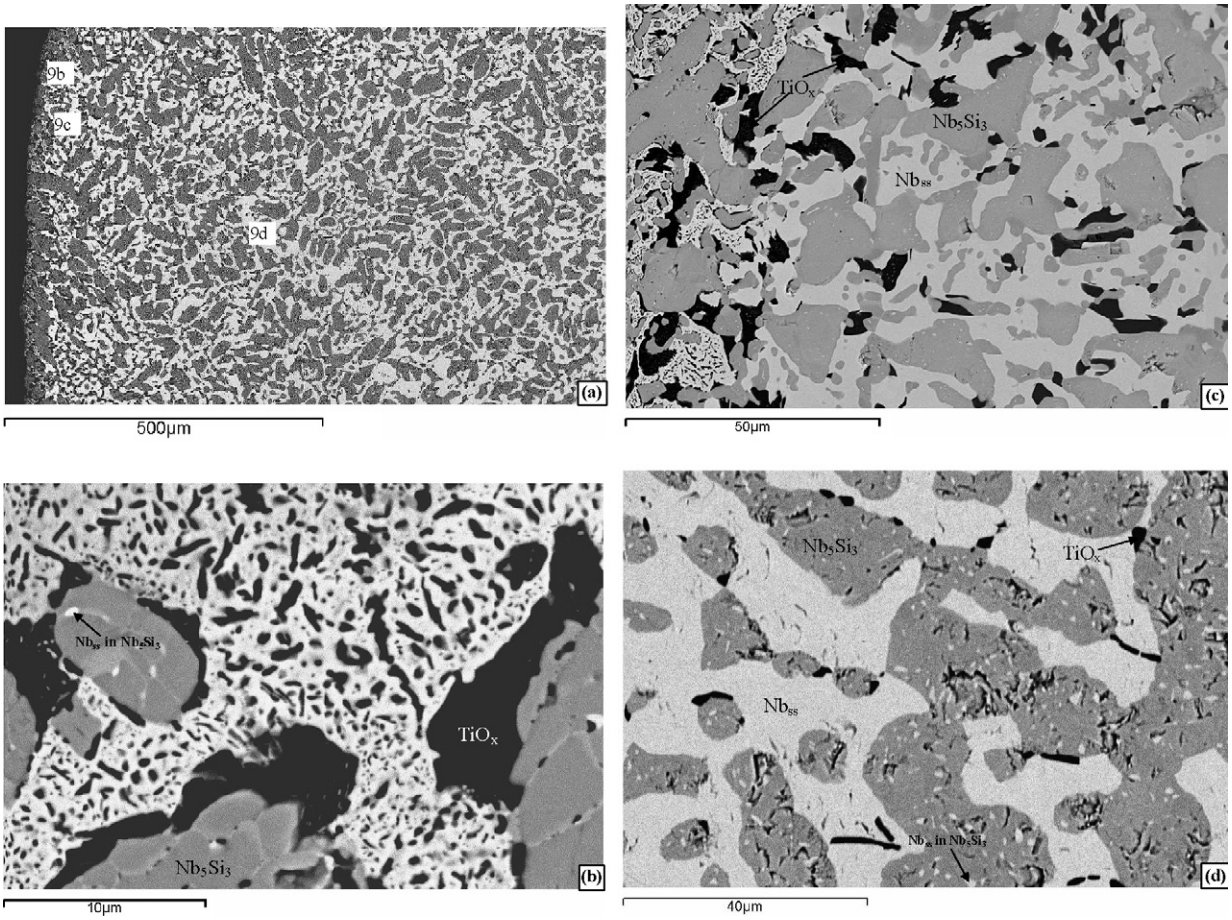


Fig. 9. BSE images of the cross section of the oxidised JG3-HT specimen at 1200 °C: (a) overall view; (b) diffusion zone I; (c) diffusion zone II; and (d) core of the specimen. The black phase is titanium oxide.

5–3 silicide, sometimes exhibited a lamellar structure. It should be noted (i) that only a small amount of needle-like HfO_2 formed in the oxidised JG4-AC, compared to the oxidised JG4-HT and (ii) that Hf-rich regions were present in the 5–3 silicides at some of their interfaces with the solid solution in both the as-cast and heat-treated conditions [18]. Thus, it is likely that the needle-like HfO_2 was formed in the prior Hf-rich areas of the 5–3 silicide.

In the diffusion zone II (Fig. 11c), the microstructure of oxidised JG4-HT consisted of four phases, the Nb_{ss}, 5–3 silicide, HfO_2 and TiO_x. Like the diffusion zone I, shown in Fig. 11b, the HfO_2 was found to be either faceted/bulky or needle-like. However, in the oxidised JG4-AC (Fig. 10c), the titanium oxide particles were absent in the diffusion zone II, there was no Hf oxide formed in the core of the oxidised JG4-AC (Fig. 10d) and no needle-like HfO_2 was observed in the diffusion zone II in the oxidised JG4-AC, suggesting that Hf exercised some control over the diffusion of oxygen towards the core of the specimen during oxidation at 1200 °C by forming the bulky HfO_2 particles in diffusion zone I.

The microstructure of the core of the oxidised specimen of JG4-AC (Fig. 10d) consisted only of the Nb_{ss} and 5–3 silicide. However, the microstructure of the core of the oxidised JG4-HT (Fig. 11d) contained three phases, the Nb_{ss}, 5–3 silicide (which exhibited cracking), and HfO_2. The needle-like HfO_2 was formed inside the 5–3 silicide. Under BS conditions in the SEM, there were various contrasts at the interface between the Nb_{ss} and 5–3 silicide. The dark region in the 5–3 silicide had lower Hf content.

Table 6 gives the compositions of phases present in the JG4-HT and the core of the oxidised JG4-HT. No Hf was detected in the Nb_{ss} after oxidation. However, the concentrations of the other solutes in the Nb_{ss} were almost unchanged compared to the heat-treated condition.

4. Discussion

The results of this work show that the choice of alloying elements and their concentrations, their partitioning behaviour as well as the types and compositions of phases present in the as-cast and heat-treated microstructures can have significant effects on the oxidation behaviour of Nb silicide-based *in situ* composites. The oxidation of all the alloys studied in this work is inferior to Ni-based superalloys. As will be discussed below, this unsatisfactory oxidation behaviour is attributed to the formation of a poorly adherent and cracked oxide scale, an oxidation zone of complex composition and microstructure and the sensitivity of the alloys to oxygen intake.

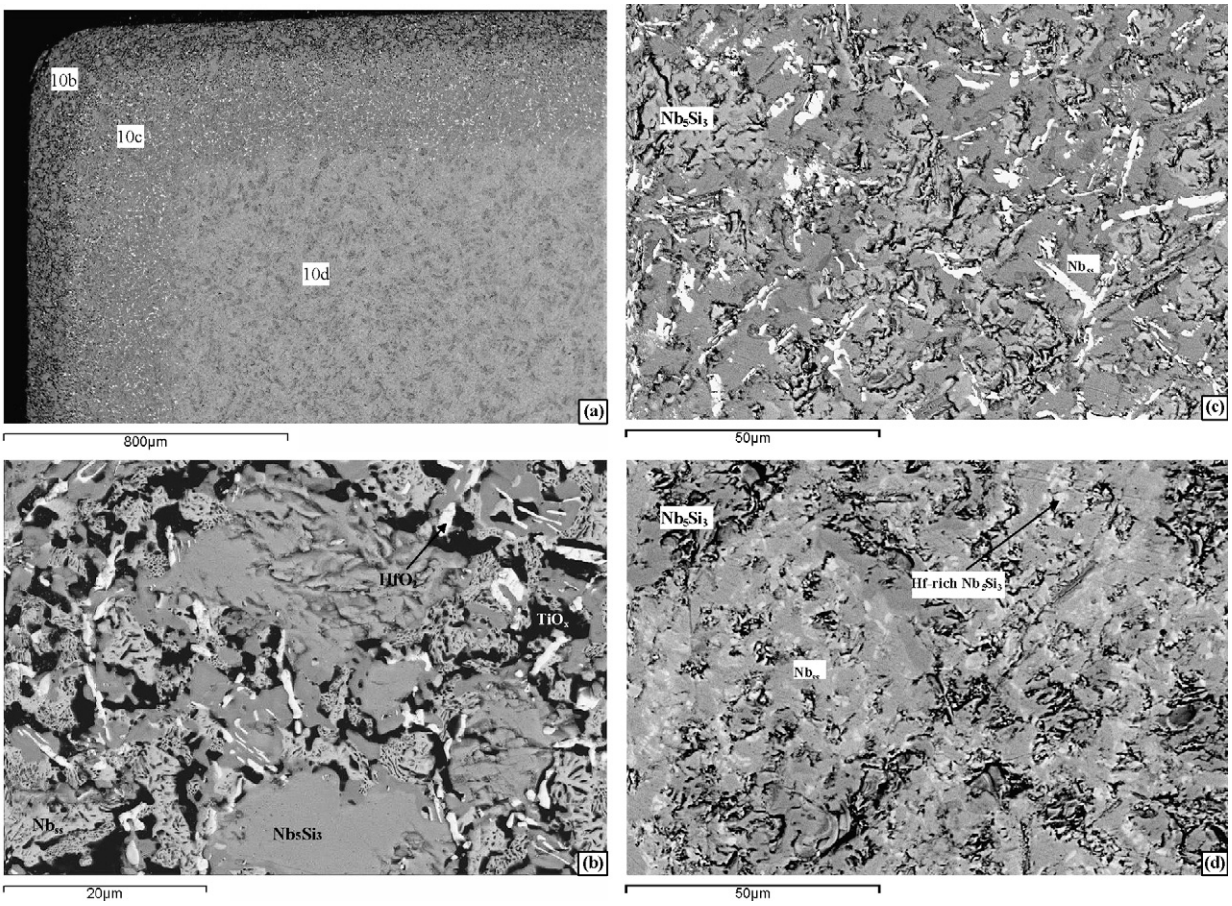


Fig. 10. BSE images of the cross section of the oxidised JG4-AC specimen at 1200 °C: (a) overall view; (b) diffusion zone I; (c) diffusion zone II; and (d) core of the specimen. The black phase is titanium oxide and the white phase is HfO_2.

The addition of Ti to JG1-AC, which suppressed the formation of the Laves phase in JG2-AC and reduced the volume fraction of Nb_{ss} by 15.6% (Table 1), was not accompanied by a change in oxidation kinetics at 800 °C (Table 2) but brought a reduction in the linear oxidation rate by 70%, and confirmed: (i) the crucial role played by the solid solution in the oxidation of Nb silicide-based *in situ* composites and (ii) that significant improvements in oxidation rate can be achieved without the presence of Laves phase in the alloy microstructure. The linear oxidation kinetics of the alloys would suggest that as the oxide scale thickened, it developed cracks or lifted away from the alloy partially or completely, and that it did not provide an effective barrier to oxygen ingress towards the alloy as oxidation progressed.

The reduction in the concentration of Mo in JG3-AC compared to JG2-AC brought a change in oxidation kinetics at 800 °C from linear to parabolic and a further improvement in oxidation behaviour (Fig. 3) even though there was a 5% increase in the volume fraction of Nb_{ss}, thus confirming that the actual concentration of Mo in the alloy does play a key role in the oxidation of Nb silicide-based *in situ* composites. The addition of Hf in JG4-AC did not change the oxidation kinetics at 800 °C but the parabolic oxidation rate constant increased by ∼90%. In JG4-AC the HfO_2 was not found in the oxide scale and diffusion zone. In JG4-AC the Nb_3Si silicide was not formed compared to JG1-AC, JG2-AC and JG3-AC, the volume fraction of the Nb_{ss} increased by ∼11% compared to JG3-AC and the Laves phase was present, thus again confirming the key role of the Nb_{ss} in the oxidation of the alloys.

In the heat-treated JG3 (no Hf present) and JG4 (Hf present) alloys, where the volume fractions of the Nb_{ss} had increased by 31.5% and 30.6%, respectively, compared to the as-cast alloys, there was a change of the oxidation kinetics at 800 °C from parabolic to linear and the linear oxidation rate constant of JG-4HT was increased by 25% compared to JG3-HT. Furthermore, the linear oxidation rate constants of JG3-HT and JG4-HT at 800 °C had increased by 40% and 75%, respectively, compared to JG2-AC, but were still lower than JG1-AC. The data for the heat-treated alloys pointed out that the beneficial effect of the reduction of the Mo concentration on the oxidation behaviour of the as-cast alloys is lost by heat treatment, and confirmed the dominance of the Nb_{ss} and the marginal role of Hf in the oxidation behaviour of the alloys at 800 °C.

Addition of Hf in JG4-AC reduced the linear oxidation rate constant by 45% after 56 h isothermal oxidation at 1200 °C compared to JG3-AC. For the heat-treated JG4 there was a change in oxidation kinetics from linear to parabolic after 24 h at 1200 °C, by which time there was a reduction in the linear oxidation rate constant for JG3-HT. Since the increase in the volume fraction of Nb_{ss} in JG3-HT and JG4-HT was almost the same (∼31%)

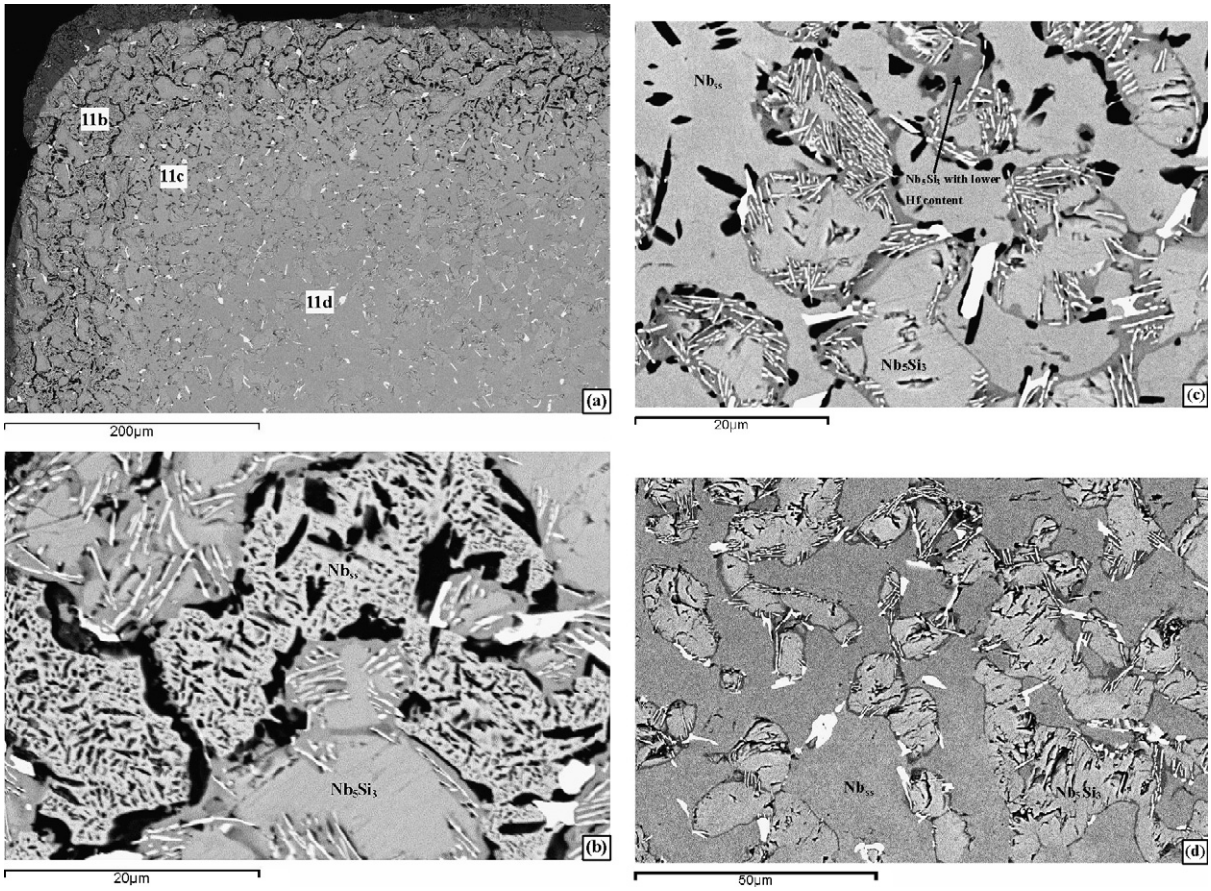


Fig. 11. BSE images of the cross section of the oxidised JG4-HT specimen at 1200 °C: (a) overall view; (b) diffusion zone I; (c) diffusion zone II; and (d) core of the specimen. The black phase is titanium oxide and the white phase is HfO_2.

compared to the as-cast alloys, and the volume fraction of Nb_{ss} in JG4-HT was higher than in JG3-HT, the data for the isothermal oxidation of JG3-HT and JG4-HT at 1200 °C would suggest that Hf might play a more important role in the oxidation of the alloys at this temperature compared to 800 °C.

In this work, pesting oxidation behaviour was observed for the JG1-AC, JG2-AC, JG3-HT and JG4-HT alloys at 800 °C. The first alloy disintegrated completely while pest-like oxidation product (powder) was formed on the oxidised specimens of the other three alloys but this pesting behaviour was not catastrophic as only part of the specimen was converted into powder after 100 h at 800 °C. However, the JG3 and JG4 alloys did not pest at 800 °C in their as-cast condition and at 1200 °C in their as-cast and heat-treated conditions. At 800 °C, oxidation scale and oxygen diffusion zone were formed in the surfaces of JG3-AC and JG4-AC. There was also severe cracking in the scale, particularly of the 5–3 silicide, with the cracks being parallel to the surface.

Pest oxidation is a dramatic oxidation phenomenon observed in intermetallic alloys and leads to disintegration of a susceptible compound or alloy into small particles or powder [20]. The Nb_3Si and Nb_5Si_3 intermetallics suffer from pest oxidation at 800 °C and 1200 °C in static air, where they disintegrate completely into powder in less than 24 h [21]. The Nb–24Ti–18Si alloy pests at 800 °C but additions of Cr or Al either alone or together, each at 5 at.% concentration, change the oxidation behaviour from pesting to one where strongly or weakly adhering thick or thin scales are formed on the base alloy [22].

Table 6
Compositions (at.%) of the phases present in JG4-HT and the core of oxidised JG4-HT (1200 °C/100 h)

Alloy	Phase	O	Al	Si	Ti	Cr	Nb	Mo	Hf
JG4-HT	Nb_{ss}	–	7.1	0.2	26.1	9.3	52.0	3.4	1.9
	Nb_5Si_3	–	2.2	35.4	17.1	0.4	40.3	0.0	4.6
	Hf-rich Nb_5Si_3	–	3.1	35.4	24.0	1.1	27.2	0.0	9.2
Oxidised JG4-HT	Nb_{ss}	0.0	7.0	0.0	25.9	9.7	54.0	3.4	0.0
	Nb_5Si_3	0.0	3.2	35.2	23.8	0.9	27.5	0.0	9.4
	HfO_2	71.3	0.0	0.0	0.0	0.0	0.0	0.0	28.7

The accuracy of oxygen analysis was about ±4.5 at.%.

The reaction of Nb with oxygen leads to the dissolution of oxygen in the metal lattice and the formation of sub-oxide phases and several homogeneous crystalline oxides in various oxidation states. At 800 °C and 1200 °C the diffusion distances of oxygen in Nb after 100 h are about 2.75 and 15.8 mm, respectively, and higher than those of nitrogen, which are 0.6 and 6.2 mm for the two temperatures, respectively [23]. Titanium reduces the diffusivity of oxygen in Nb, for example, for Nb–25Ti the diffusivity is 1/20 that in pure Nb [24].

In our alloys, as oxygen anion diffusion occurred towards their bulk, tensile and compressive stresses developed due to the change of oxide composition and the formation of NbO_x ($x < 1$), NbO_y ($y = 1, 2$) and Nb_2O_5 oxides with Pilling Bedworth ratios less than one (NbO_x) or greater than one (NbO_y ($y = 1, 2$) and Nb_2O_5). Furthermore, tensile and compressive stresses developed between different regions of Nb_{ss} with different oxygen content [25] and because of phase transformations of the NbO_2 at ~770 °C [26] and Nb_2O_5 at ~800 °C in the absence of Ti, or at lower temperatures (~700 °C) in its presence, as Ti (and Cr) extend the stability of the high temperature Nb_2O_5 to lower temperatures whilst Si stabilises the low temperature form of Nb_2O_5. A decrease in the oxidation rate of Nb–Ti alloys at ~700 °C has been attributed by Argent and Phelps to different diffusion rates of oxygen in the low and high temperature forms of Nb_2O_5 [27]. A sharp decrease in the oxidation rate of Nb–Si alloys at 900 °C has also been reported by Magnelli et al. [28], who attributed this change to the transformation of Nb_2O_5 from the low to the high temperature form.

The presence of Nb_2O_5, TiO_2 and SiO_2 in the oxide scale formed on JG4-AC and of Nb_2O_5, TiO_2, $TiNb_2O_7$ and SiO_2 in the oxide scale formed on JG3-AC was indicated by the XRD data for 800 °C (see Fig. 4). According to Roth and Coughanour [29], TiO_2 and Nb_2O_5 form solid solutions and the compounds $3Nb_2O_5$–TiO_2 ($T_{mp} \sim 1480$ °C) and Nb_2O_5–TiO_2 ($T_{mp} \sim 1490$ °C). Goldschmidt [30] has suggested: (i) that the $3Nb_2O_5$–TiO_2 is not a special compound but instead is a solid solution based on the high temperature form of Nb_2O_5 and (ii) that the Nb_2O_5–TiO_2 compound can be regarded as an independent phase with a considerable homogeneity range and a monoclinic structure that has lattice parameters different from those of the high temperature form of Nb_2O_5 [30,31]. Wadsley has proposed that instead of $3Nb_2O_5$–TiO_2 the compound $5Nb_2O_5$–TiO_2 (or $Ti_2Nb_{10}O_{29}$) exists in two forms, at low temperatures with orthorhombic structure and at high temperatures with monoclinic structure. The presence of $Ti_2Nb_{10}O_{29}$ is suggested by the EPMA and XRD data (see Table 5, Figs. 4 and 7) for the oxide scales formed on JG3 and JG4 at 1200 °C but not at 800 °C.

Fig. 5 shows that the Nb_5Si_3 phase, both in the diffusion zone or in the core of the base materials, remained un-attacked after isothermal oxidation tests at 800 °C, and thus confirm that preferential attack of the Nb_{ss} phase had taken place in our alloys at 800 °C.

Changes in growth rate/s accompanied by the phase transformations of the oxide discussed above led to the development of stresses in the growing oxide scale. Furthermore, lateral oxide growth developed compressive stresses that caused local debonding when the stresses exceeded a critical value [32]. The growth stresses that developed either from a single mechanism or from a combination of different mechanisms, as discussed above, induced microcracks in the growing oxide scale, and caused cracking of the Nb_5Si_3 phase, which, as can be seen in Fig. 5, exhibited severe cracking. These cracks in the 5–3 silicide provided more sites for rapid oxygen ingress, growth of voluminous oxides and stress risers, as discussed above. The reaction then continued on the newly exposed surfaces and within newly exposed cracks and pores, eventually leading to full disintegration of the alloy.

A protective Al_2O_3 or SiO_2 scale was not formed at 800 °C or 1200 °C in any of the alloys studied in this work. The accelerated oxidation of JG1-AC that led to the disintegration of the alloy to powder is attributed to rapid oxygen ingress and internal oxidation by transport through non-protective Nb oxide product(s) and the formation of an oxygen-affected zone that led to a mixture of oxides or oxygen-saturated metal + oxides being formed. It is suggested that only after such a microstructure/morphology of the scale was established, that the characteristic pest disintegration product (powder) occurred. It resulted from the action of stresses from the formation of fast-growing voluminous oxides. Such stresses contributed to the breakdown of any partially protective reaction products formed during the preceding accelerated oxidation stage in the case of JG1-AC but not for JG2-AC and JG3 and JG4 in their as-cast and heat-treated conditions.

The need to establish a susceptible microstructure would account for the behaviour of JG3-HT and JG4-HT. Accelerated oxidation alone was not sufficient for pesting in our alloys; oxide growth stresses in the surfaces of the alloys, which have limited ductility and toughness [2] had also to be present and these had to be above the stress levels that led to the cracking of the 5–3 silicides in JG3-AC and JG4-AC. Thus, the pesting oxidation behaviour of our alloys occurred due to a combination of the presence of elements that oxidised rapidly and the restricted deformation capability of the scale.

The pesting of JG1-AC and JG2-AC at 800 °C would suggest that the 5 at.% Mo concentration in these alloys, which leads to 8–9 at.% Mo in their Nb_{ss} and 1.5–2.5 at.% Mo and 3.5–4.5 at.% Mo in the 5–3 and 3-1 silicides, respectively [17], is high regarding avoidance of pest oxidation. The oxidation behaviour of JG3 and JG4 at 800 °C and 1200 °C in their as-cast and heat-treated conditions would suggest that the Mo concentration in the alloy should not exceed 2 at.% for oxidation resistance purposes.

In the as-cast and heat-treated conditions the Nb_{ss} and 5–3 silicides, which were present in the microstructures of JG3 and JG4, had Ti and/or Hf-rich regions [17,18]. Furthermore, TiN had formed near the surfaces of specimens that had been heat-treated at 1500 °C for 100 h or 200 h. The TiN phase was not found in the alloys oxidised at 800 °C and 1200 °C. The same oxides were formed on JG3 and JG4, with HfO_2 being also present in the scale formed on the latter alloy. As discussed above, the $Ti_2Nb_{10}O_{29}$ undergoes a phase transformation on heating from a low temperature orthorhombic structure to a high temperature monoclinic structure [31]. The NbO_2 and TiO_2

oxides form solid solutions and the $TiNbO_4$ compound [30]. The oxidation of pure Nb from lower (NbO) to higher (Nb_2O_5) oxides is accompanied by an increase in relative volume, with the lower oxides forming on the base metal a dark film and the higher oxide (Nb_2O_5) undergoing a phase transformation at $T > 800\,^{\circ}C$ (see above). In Nb–Ti alloys and at $T \geq 1150\,^{\circ}C$ the oxidation rate of Nb_2O_5 increases further due to the evolution of large amount of heat [27]. The high temperature oxidation of Nb_5Si_3 (Nb_2O_5 is the oxidation product at 1200 °C) [21] attains a maximum at ∼1100 °C and then drops in the range 1200–1400 °C [33].

The combined effects: (i) of Ti, Cr and Si on the phase transformation of Nb_2O_5 and its compounds (see above), (ii) of the mechanisms responsible for the damage of the oxide scale formed on the alloys (see above) and (iii) of the above mechanisms operating either alone or in synergy at $T > 800\,^{\circ}C$ led to further growth stresses developing in the oxidation scale formed at 1200 °C, mechanical damage (cracking) of the scale and further oxygen anion access to the underlying alloy leading to formation of $(Nb, O)_{ss}$, TiO_2 and HfO_2 (in JG4) in the regions below the diffusion zone II. The cracked and porous oxide scale together with the stresses developing during cooling of the oxidised specimens led to the spallation of the scales formed on JG3 and JG4.

The microstructures of the diffusion zones developed in JG3 and JG4 after exposure at 1200 °C (Figs. 9–11) exhibited: (i) fine titanium oxide particles inside the niobium solid solution and (ii) large titanium oxide particles at the interfaces between the Nb_{ss} and the 5–3 silicide. Furthermore, in the case of the JG3-HT oxidised at 1200 °C the Ti-rich 5–3 silicide was absent in the microstructures of the diffusion zones and the underlying alloy, with titanium oxide being present in the latter (Fig. 9d) and the Nb_{ss} particles inside the 5–3 silicide [17] appearing intact. Considering these results, it is suggested that oxygen, which had diffused in the Nb_{ss}, reacted with Ti in the Ti-rich 5–3 silicide to form Ti oxide particles at the interface. The formation of Ti oxide at the interface between Nb_{ss} and 5–3 silicide and of fine Ti oxide particles inside the Nb_{ss} is thus important regarding the oxidation behaviour of the JG3 and JG4 alloys at 1200 °C.

The needle-like HfO_2 was formed inside the 5–3 silicide in JG4-HT, but not in JG4-AC after isothermal exposure at 1200 °C (see Figs. 10 and 11). As discussed in the Section 3.2, the Hf-containing alloy JG4 showed better oxidation resistance in the as-cast condition but this behaviour was reversed after heat treatment, compared to the Hf-free alloy JG3. The formation of needle-like HfO_2 inside the 5–3 silicide in the oxidised JG4-HT had contributed to the degradation in the oxidation resistance of the Hf-containing alloy in the heat-treated condition.

The EPMA results (Table 6) indicated that no Hf was present in the Nb_{ss} phase in the core of the oxidised JG4-HT. This probably resulted from the formation of faceted HfO_2, which was formed inside the Nb_{ss}, as shown in Fig. 9c. The EPMA data also showed that the composition of the Hf-rich 5–3 silicide had not changed after oxidation at 1200 °C, compared to that of the heat-treated alloy JG4HT [18], which suggests that not all the 5–3 silicide areas had been attacked by oxygen to form HfO_2.

5. Conclusions

The oxidation behaviour of four Nb–Si–Cr–Al-based *in situ* composites in the as-cast and heat-treated conditions was studied to understand the effects of the additions of Hf, Mo and Ti on oxidation at 800 °C and 1200 °C. The conclusions of this study are as follows.

1. The oxidation resistance of the Nb–Si–Cr–Al silicide-based *in situ* composites at 800 °C was dramatically increased by the addition of Ti and the decrease of the Mo content in the alloy. In the presence of Ti, Al, Cr and Mo, the Hf addition had no significant effect on the oxidation behaviour of the Nb silicide-based *in situ* composites at 800 °C but was probably beneficial to oxidation at 1200 °C. No protective oxide scale was formed on any of the alloys studied in this work. Oxidation-protective coatings, therefore, will have to be used for the application of these alloys in oxidising environments.
2. The oxidation behaviour of all the alloys at 800 °C and 1200 °C was controlled by the oxidation of the Nb_{ss} and was sensitive to the volume fraction of Nb_{ss} in the alloy. Preferential attack of the Nb_{ss} phase was observed in all the oxidised alloys.
3. Pesting oxidation behaviour was exhibited by the Nb–18Si–5Al–5Cr–5Mo (as-cast), Nb–24Ti–18Si–5Al–5Cr–5Mo (as-cast), Nb–24Ti–18Si–5Al–5Cr–2Mo (heat-treated) and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf (heat-treated) alloys at 800 °C. Pesting of the alloys occurred due to a combination of the presence of elements that oxidised rapidly and the restricted deformation capability of the scales.
4. Spallation of oxide scales during cooling was observed for the as-cast and heat-treated Nb–24Ti–18Si–5Al–5Cr–2Mo and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf alloys at 1200 °C. The oxides $TiNbO_4$, $TiNb_2O_7$, $Ti_2Nb_{10}O_{29}$ and silicon oxide were present in all the scales formed on both the Nb–24Ti–18Si–5Al–5Cr–2Mo and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf alloys at 1200 °C and Hf oxide was also present in the latter alloy.
5. The oxidation behaviour of the Nb–24Ti–18Si–5Al–5Cr–2Mo and Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf alloys at 800 °C and 1200 °C in their as-cast and heat-treated conditions suggests that the Mo concentration in the alloy should not exceed 2 at.% for oxidation resistance purposes. At 1200 °C the two alloys formed large Ti oxide particles at the interfaces between Nb_{ss} and 5–3 silicides, fine Ti oxide particles inside the Nb_{ss} and a two-zone microstructure in the diffusion zones that developed on each alloy.
6. In the diffusion zones formed in the alloys without Hf addition, the 5–3 silicide was not oxidised. However, needle-like Hf oxide formed inside the 5–3 silicide in the heat-treated Nb–24Ti–18Si–5Al–5Cr–2Mo–5Hf alloy.

Acknowledgement

Financial support to Jie Geng by the University of Surrey is gratefully acknowledged.

References

[1] M.R. Jackson, B.P. Bewlay, R.G. Rowe, D.W. Skelly, H.A. Lipsitt, JOM 48 (1) (1996) 39–44.
[2] B.P. Bewlay, M.R. Jackson, J.C. Zhao, P.R. Subramanian, Metall. Mater. Trans. A 34A (2003) 2043–2052.
[3] S. Jiangba, Mater. Sci. Eng. A 343A (2003) 282–289.
[4] M.G. Mendiratta, D.M. Dimiduk, Metall. Trans. A 24A (1993) 501–504.
[5] M.G. Mendiratta, J.J. Lewandowski, D.M. Dimiduk, Metall. Trans. A 22A (1991) 1573–1583.
[6] M.R. Jackson, B.P. Bewlay, J.C. Zhao, R.R. Corderman, US Patent 6428910 (2002).
[7] P.R. Subramanian, M.G. Mendiratta, D.M. Dimiduk, JOM 48 (1) (1996) 33–38.
[8] P.R. Subramanian, M.G. Mendiratta, D.M. Dimiduk, M.A. Stucke, Mater. Sci. Eng. A 239–240A (1997) 1–13.
[9] B.P. Bewlay, M.R. Jackson, H.A. Lipsitt, Metall. Mater. Trans. A 27A (1996) 3801–3808.
[10] K.S. Chan, Mater. Sci. Eng. A 329–331A (2002) 513–522.
[11] K.S. Chan, D.L. Davidson, Metall. Mater. Trans. A 34A (2003) 1833–1849.
[12] B.P. Bewlay, C.L. Briant, A.W. Davis, M.R. Jackson, MRS Symp. Proc. 646 (2001) N2.7.1–N2.7.6.
[13] R.T. Begley, in: E.N.C. Dadler, T. Grobstein, C.S. Olsen (Eds.), Evolution of Refractory Metals and Alloys, TMS, Warrendale, 1994, pp. 29–48.
[14] B.P. Bewlay, M.R. Jackson, M.F.X. Gigliotti, in: J.H. Westbrook, R.L. Fleischer (Eds.), Intermetallic Compounds: Principles and Practice, vol. 3, John Wiley & Sons, London, 2002, pp. 541–560.
[15] G.D. McAdam, J. Inst. Met. 93 (1964–1965) 559–564.
[16] J.-H. Kim, T. Tabaru, M. Sakamoto, S. Hanada, Mater. Sci. Eng. A. 372 (2004) 137–144.
[17] J. Geng, P. Tsakiropoulos, G. Shao, Intermetallics 14 (3) (2006) 227.
[18] J. Geng, P. Tsakiropoulos, G. Shao, Intermetallics, doi:10.1016/j.msea.2006.08.093, in press.
[19] M.P. Brady, B.A. Pint, P.F. Tortorelli, I.G. Wright, in: M. Schutze (Ed.), Material Science and Technology: Corrosion and Environmental Degradation, vol. II, Wiley-VCH, Weinheim, Germany, 2000, p. 243.
[20] E.A. Aitken, in: J.H. Westbrook (Ed.), Intermetallic Compounds, Wiley, New York, 1967, pp. 491–516.
[21] N. Vellios, P. Tsakiropoulos, Metallurgy Research Group, University of Surrey, unpublished research.
[22] K. Zelenitsas, P. Tsakiropoulos, Mater. Sci. Eng. A. 416 (2006) 264–280.
[23] G.J. Ang, Acta Metall. 1 (1953) 123–129.
[24] K.K. Kelly, Bull. Bureau Mines 476 (1949) 22–29.
[25] D.A. Prokoshkin, E.V. Vasil'eva, Alloys of Niobium, Israel Program for Scientific Translations, Jerusalem, 1965, p. 80.
[26] C. Wert, C. Zener, Phys. Rev. 76 (1949) 1169–1173.
[27] B.B. Argent, B. Phelps, J. Less Common Met. 2 (1960) 181–190.
[28] A. Magnelli, G. Andersson, G. Sundkvist, Acta Chem. Scand. 9 (1955) 1402–1408.
[29] R.S. Roth, I.W. Coughanour, J. Res. Nat. Bureau Stand. 55 (1955) 209–216.
[30] H.J. Goldschmidt, Metallurgia 373 (1960) 211–217.
[31] A.D. Wadsley, Acta Crystallogr. 6 (1961) 664–670.
[32] M. Schutze, Protective Oxide Scales and their Breakdown, Institute of Corrosion and Wiley, 1997.
[33] G. Grube, O. Kubaschewski, K.Z. Zwiauer, Electrochemistry 45 (1939) 885–891.